

Quadratic function

A **quadratic function** has the form

$$f(x) = ax^2 + bx + c, \quad a \neq 0,$$

where a , b , and c are real numbers. Its graph is a parabola with a vertical axis of symmetry, and its domain is $(-\infty, \infty)$.

Quadratic equation

A **quadratic equation in standard form** is

$$ax^2 + bx + c = 0, \quad a \neq 0.$$

Its solutions are the zeros of the related quadratic function and the x -coordinates of its x -intercepts.

For $f(x) = 2x^2 + 3x - 2$, identify a , b , and c .

Zero-product property

If A and B are real numbers or algebraic expressions and

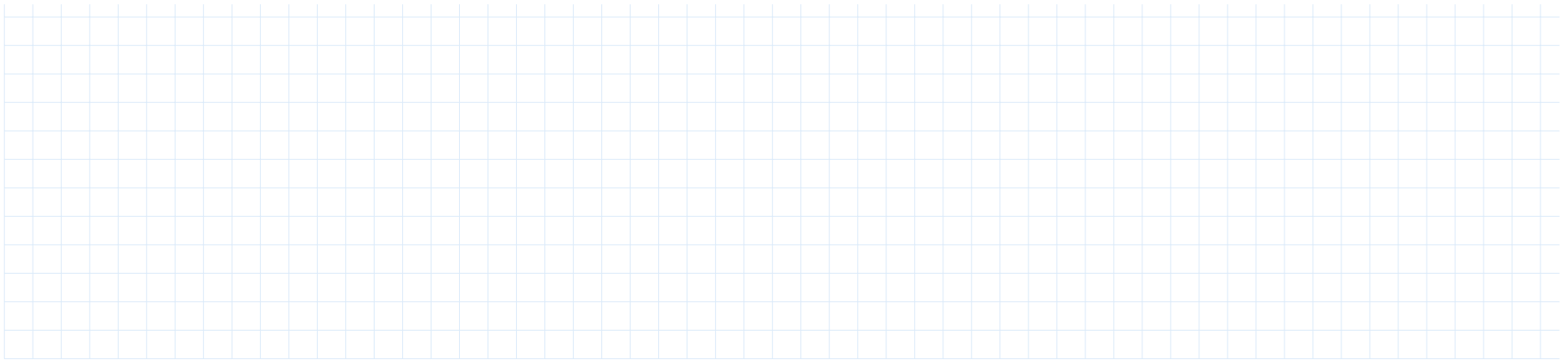
$$AB = 0,$$

then $A = 0$ or $B = 0$ (or both).

Example. Solve

$$(x + 6)(2x - 3) = 0.$$

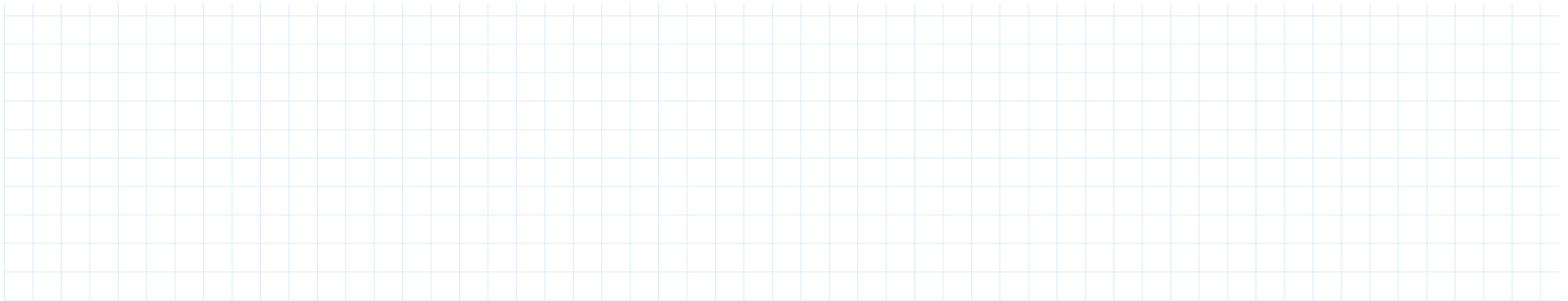
Then write the corresponding x -intercepts.



1. Write the equation in standard form.
2. Factor completely.
3. Apply the zero-product property.
4. Solve each resulting linear equation.
5. Check in the original equation.

Example. Find the zeros of

$$f(x) = 2x^2 + 3x - 2.$$

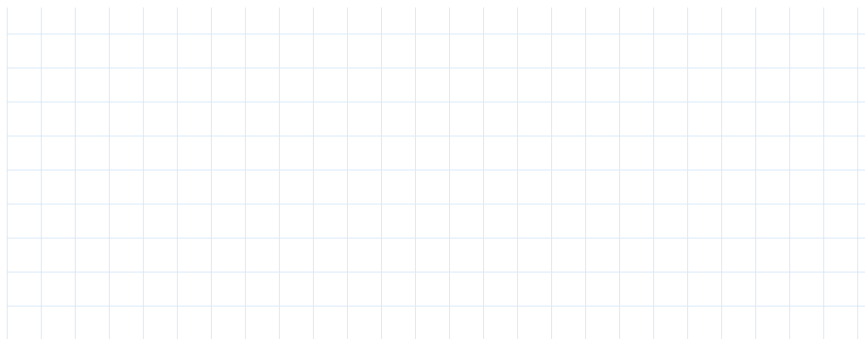


Factoring Patterns

Solve each equation by factoring. Name the pattern or common factor you use.

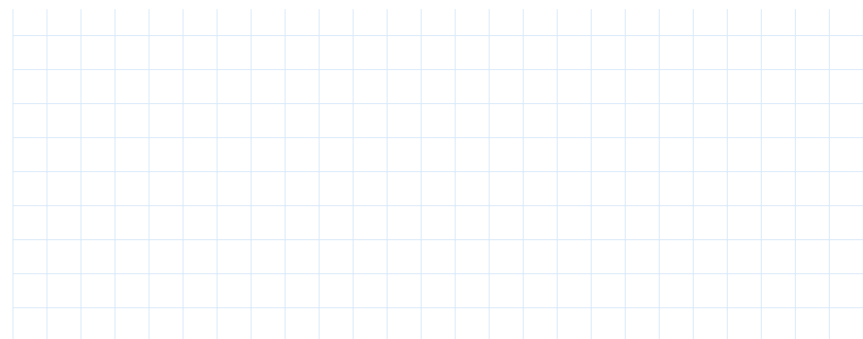
1. Common factor

$$4x^2 - 20x = 0$$



2. Difference of squares

$$3x^2 - 108 = 0$$



3. Trinomial: $2x^2 + 5x - 7 = 0$.

Square-Root Property

Square-root property

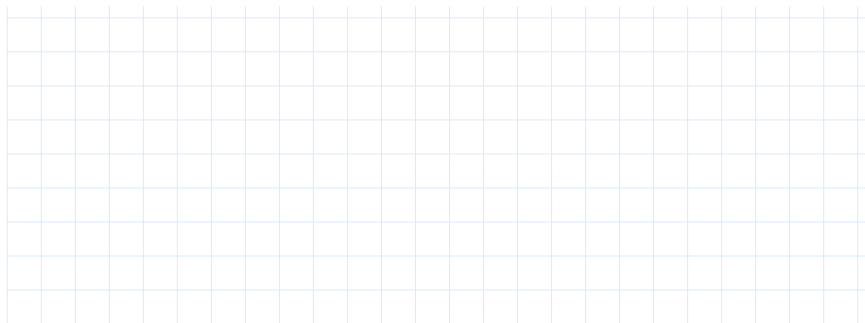
If u is an algebraic expression and $d > 0$, then

$$u^2 = d \implies u = \pm\sqrt{d}.$$

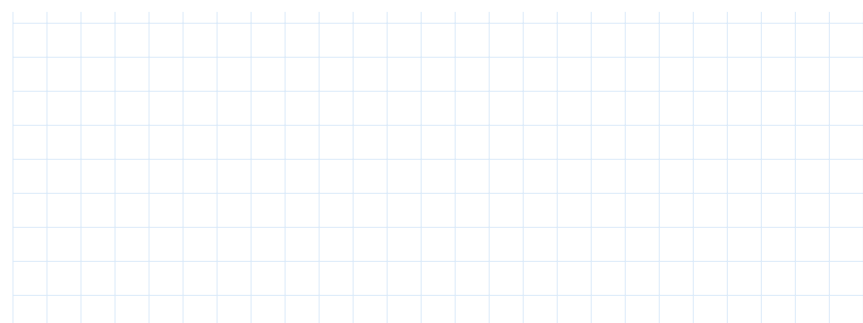
The symbol $\pm$ is necessary because both a positive and a negative number can have the same square.

Solve each equation using the square-root property.

$$x^2 - 12 = 0$$



$$(x - 2)^2 - 16 = 0$$



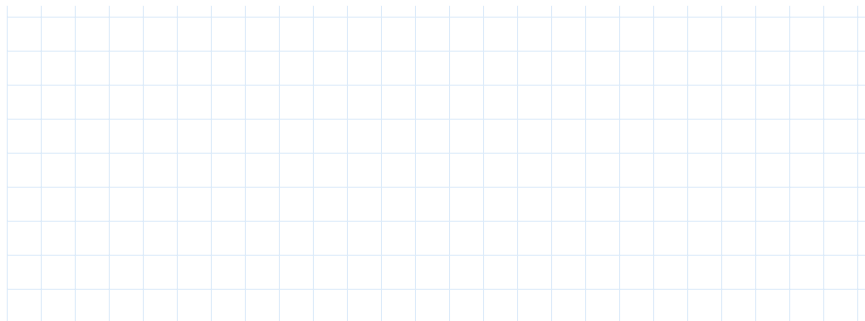
Completing the Square

For a binomial $x^2 + bx$, add the square of half the coefficient of x :

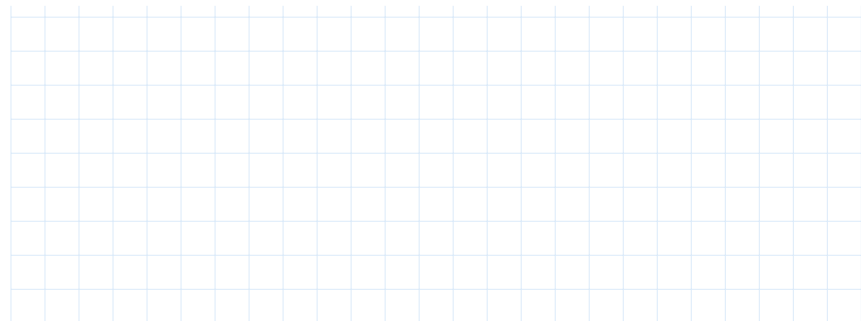
$$x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2$$

Complete each expression so that it becomes a perfect-square trinomial. Then factor it.

$$x^2 + 8x + \underline{\hspace{2cm}}$$



$$x^2 - 14x + \underline{\hspace{2cm}}$$

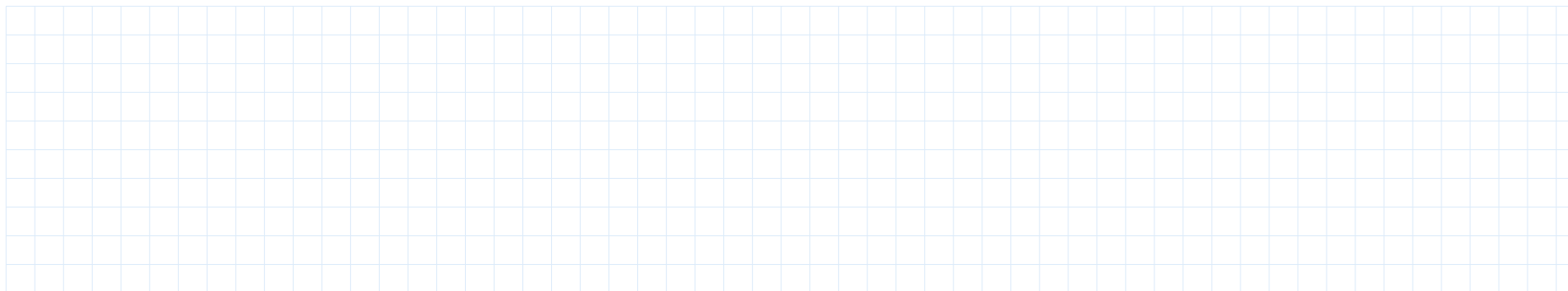


Solve by Completing the Square

To solve $ax^2 + bx + c = 0$:

1. Divide by a if $a \neq 1$.
2. Move the constant term to the other side.
3. Add $\left(\frac{b}{2}\right)^2$ to both sides.
4. Factor the perfect-square trinomial.
5. Apply the square-root property and simplify.

Example. Solve $x^2 + 8x + 10 = 0$.



Method Practice

Choose factoring, the square-root property, or completing the square. State why your method fits.

1. $x^2 + 5x - 1 = 0$

2. $2x^2 - 4x - 5 = 0$

3. $(3x + 1)^2 = 20$

